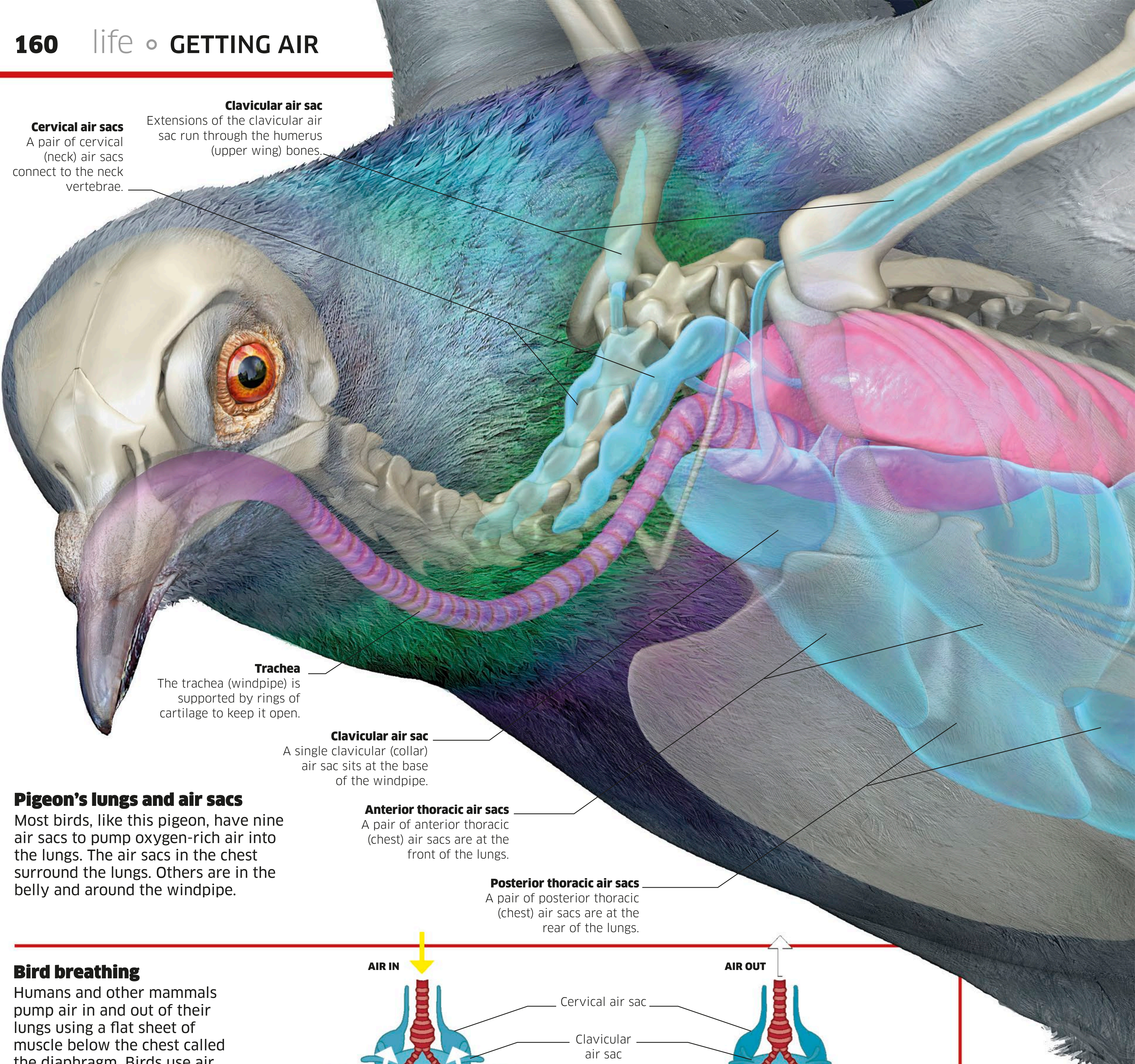
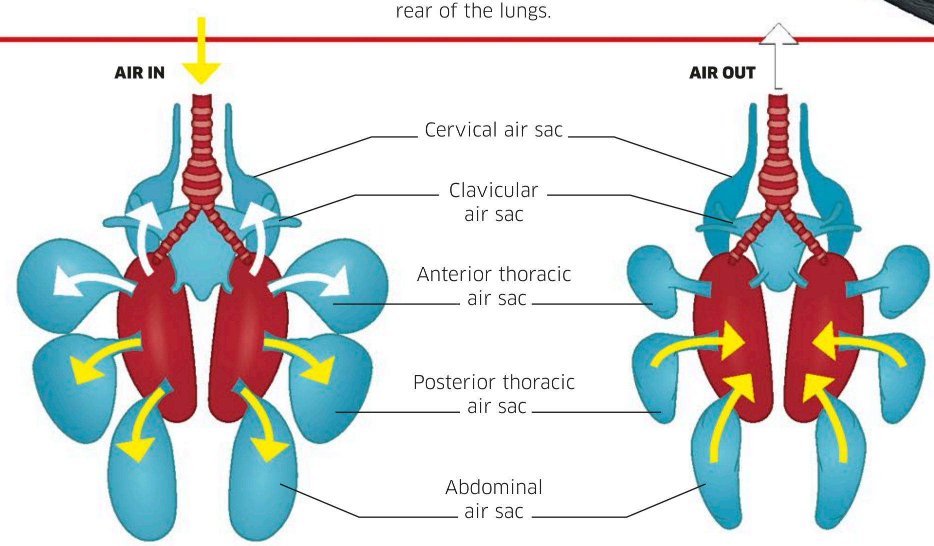


Pigeon's lungs and air sacs

Most birds, like this pigeon, have nine air sacs to pump oxygen-rich air into the lungs. The air sacs in the chest surround the lungs. Others are in the belly and around the windpipe.

Bird breathing

Humans and other mammals pump air in and out of their lungs using a flat sheet of muscle below the chest called the diaphragm. Birds use air sacs instead to move fresh air through their bodies when breathing in and breathing out. Air circulates one way from the rear air sacs, through the lungs to the front air sacs. The air sacs—which account for 20 percent of the body volume—also help to stop the bird overheating, and make swimming birds buoyant.



Inhalation
When a bird breathes in, the air entering the body first fills air sacs at the rear (yellow). At the same time, the air that was already in the lungs moves out to inflate the front air sacs (white).

Exhalation
When breathing out, all the air sacs work like bellows to pump air as they deflate. The rear air sacs fill the lungs with air, while the front air sacs push air back out through the mouth.